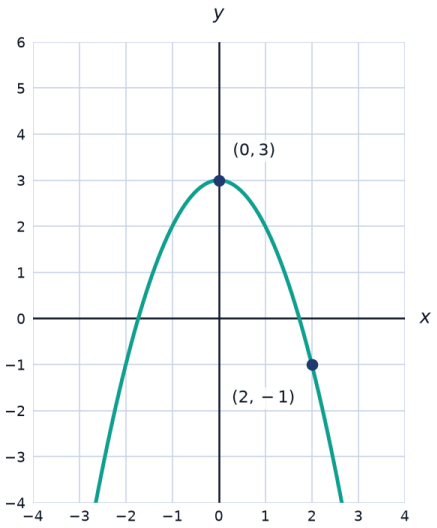
Nonlinear Functions, Function Notation & Equivalent Expressions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

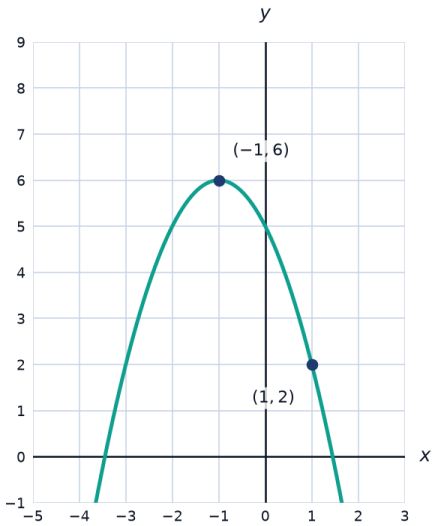
1 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the municipal library is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 4$, what is the maximum value of $g(x)$?



2 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

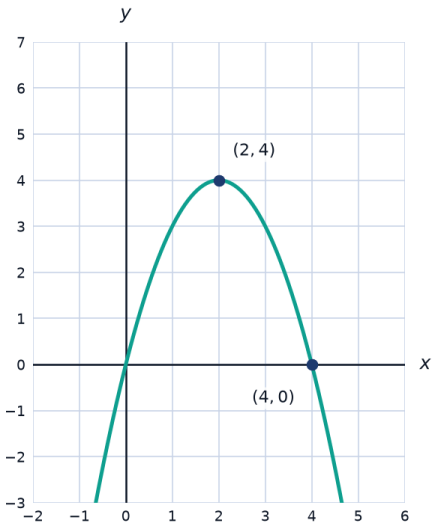
The graph of $y = f(x)$ for a sensor at the coastal aquarium is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 7$, what is the maximum value of $g(x)$?



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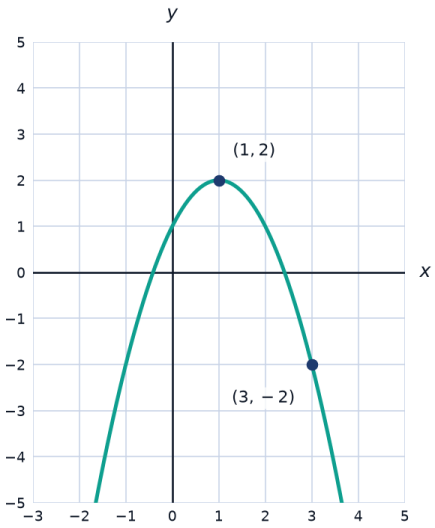
3 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the urban bike-share station is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 4$, what is the maximum value of $g(x)$?



4 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

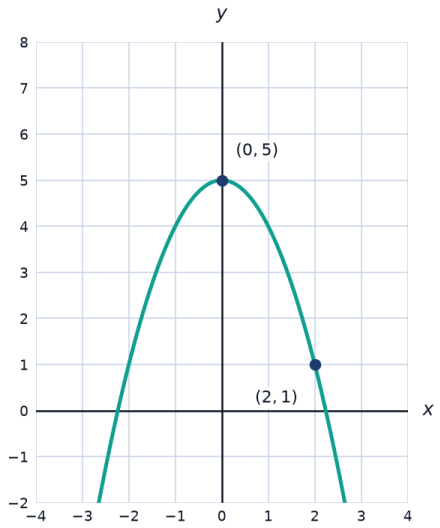
The graph of $y = f(x)$ for a sensor at the agricultural cooperative is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 7$, what is the maximum value of $g(x)$?



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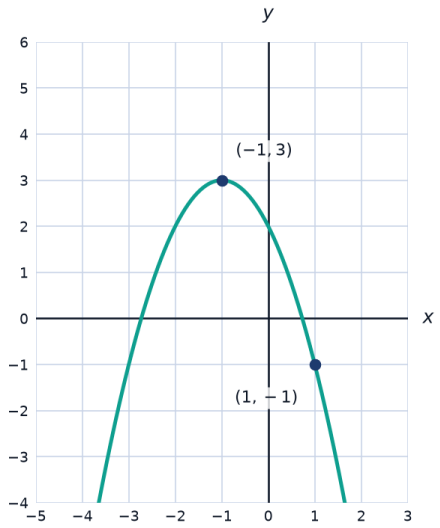
5 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the recycling center is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 4$, what is the maximum value of $g(x)$?



6 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

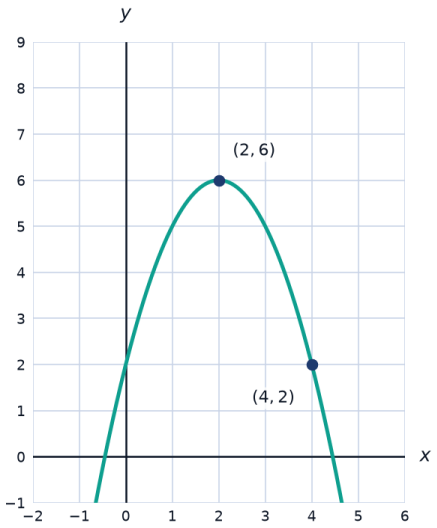
The graph of $y = f(x)$ for a sensor at the neighborhood bookstore is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 7$, what is the maximum value of $g(x)$?



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7 [Foundational] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the culinary institute is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 4$, what is the maximum value of $g(x)$?



Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the robotics club, in a calculation for the orchestra hall (case 12), if $f(x) = 2|x + 9| - 1$, what is $f(10)$?

9 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the maple syrup farm, for what value of x is the function $f(x) = \frac{1}{x - 5}$ undefined?

10 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the marine research vessel, in a calculation for the local farmers market (case 20), what is the minimum value of $f(x) = 1|x - 7| - 8$?

11 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the campus coffee roastery, in a calculation for the performing arts academy (case 24), if $f(7) = 17$, and $g(x) = f(x) - 1$, what is $g(7)$?

12 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the meteorological station, in a calculation for the coastal aquarium (case 28), if $f(-8) = -7$, and $g(x) = f(x) + 4$, what is $g(-8)$?

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13 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the urban bike-share station, in a calculation for the public transit depot (case 32), if $f(3) = -12$, and $g(x) = f(x) - 2$, what is $g(3)$?

14 [Intermediate] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the sports complex, in a calculation for the geology lab (case 40), if $h(x) = |x - 4| + 2$, what is the minimum value of $h(x)$?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the regional airport shuttle, what is the smallest value of x for which $f(x) = \sqrt{x}$ is defined?

16 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

At the botanical conservatory, the functions f and g are defined by $f(x) = 2x - 2$ and $g(x) = x^2 + 1$. What is the value of $f(g(2))$?

17 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

In the xy-plane, the graph of the quadratic function $y = f(x)$ has its vertex at $(3, 4)$. If $g(x) = f(x - 5)$ models the shifted curve at the municipal library, what is the x-coordinate of the vertex of the graph of g?

18 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

The nonlinear function $h(x) = 4x^2 + c$, where c is a constant, models the output at the community theater. If $h(4) = 71$, what is the value of $h(0)$?

19 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

At the regional bakery, the functions f and g are defined by $f(x) = 5x - 5$ and $g(x) = x^2 + 4$. What is the value of $f(g(5))$?

20 [Challenging] • [SPR] • Target: Function Evaluation, Composition, Transformations & Graphs

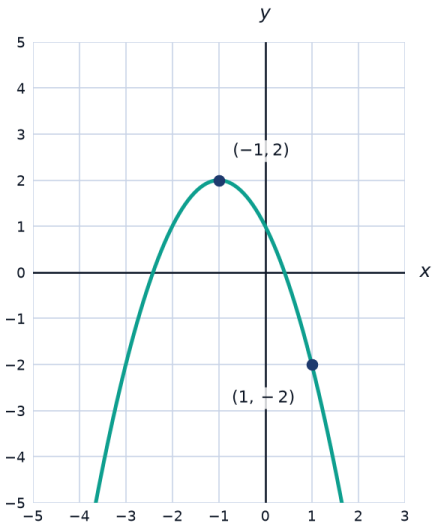
In the xy-plane, the graph of the quadratic function $y = f(x)$ has its vertex at $(6, 7)$. If $g(x) = f(x - 8)$ models the shifted curve at the coastal aquarium, what is the x-coordinate of the vertex of the graph of g?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the botanical conservatory is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 3$, what is the maximum value of $g(x)$?



a. 5

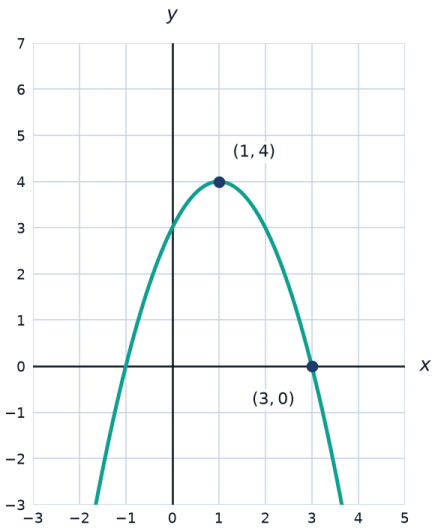
b. 2

c. -1

d. 1

22 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the community theater is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 5$, what is the maximum value of $g(x)$?



a. -1

b. 5

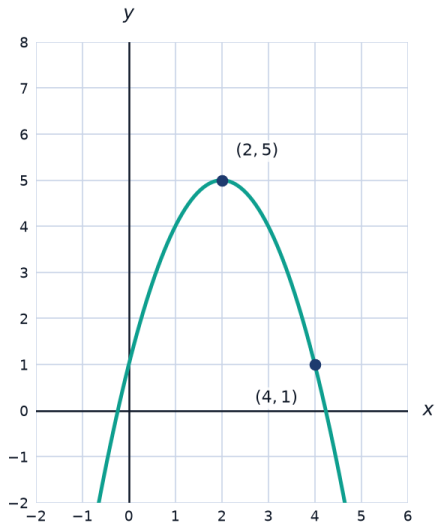
c. 9

d. 6

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23 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the regional bakery is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 6$, what is the maximum value of $g(x)$?



a. 7

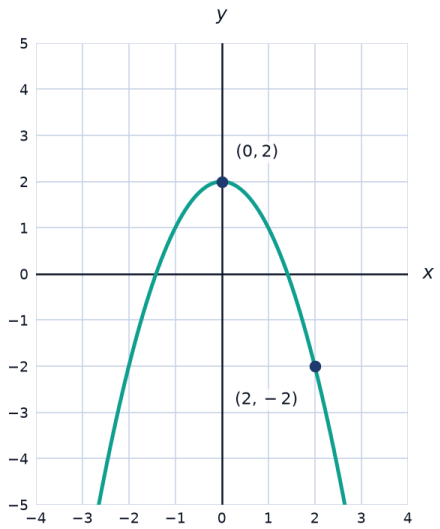
b. 11

c. 8

d. -1

24 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the mountain observatory is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 8$, what is the maximum value of $g(x)$?



a. 8

b. -6

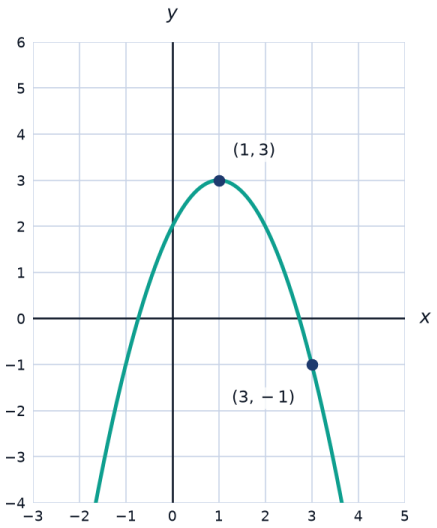
c. 6

d. 10

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25 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the science museum is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 3$, what is the maximum value of $g(x)$?



a. 0

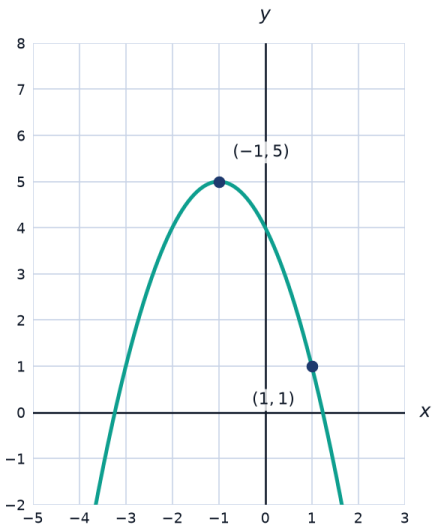
b. 2

c. 6

d. 4

26 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the solar microgrid facility is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 5$, what is the maximum value of $g(x)$?



a. 10

b. 4

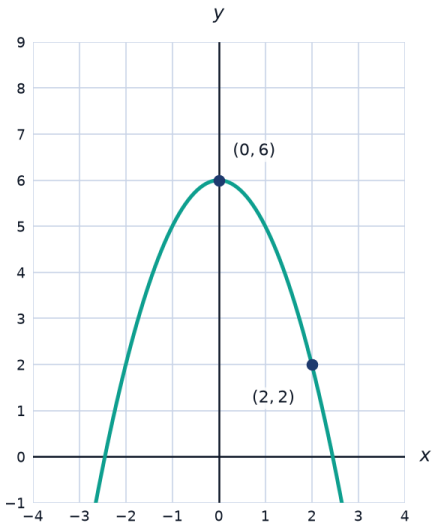
c. 0

d. 6

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27 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the ceramic studio is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 6$, what is the maximum value of $g(x)$?



a. 6

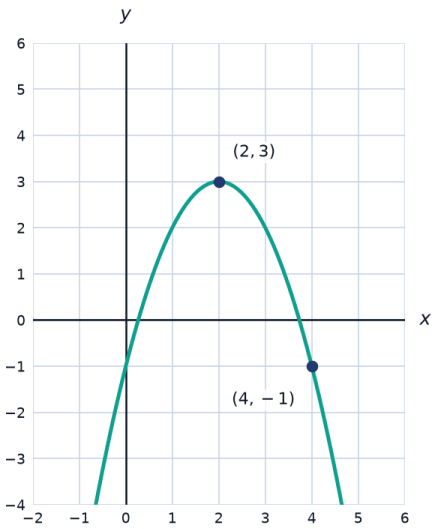
b. 0

c. 8

d. 12

28 [Foundational] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The graph of $y = f(x)$ for a sensor at the publishing house is shown in the xy-plane with its maximum point and another point labeled. If $g(x) = f(x) + 8$, what is the maximum value of $g(x)$?



a. 7

b. 11

c. 10

d. −5

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Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 **[Intermediate]** • **[MCQ]** • **Target: Function Evaluation, Composition, Transformations & Graphs**

In a model for the art gallery, in a calculation for the wildlife sanctuary (case 11), if $f(x) = 1|x + 7| - 8$, what is $f(-10)$?

a. -9

b. -5

c. -6

d. 4

30 **[Intermediate]** • **[MCQ]** • **Target: Function Evaluation, Composition, Transformations & Graphs**

In a model for the culinary institute, in the coordinate plane, the graph of $y = F(w)$ is translated 6 units right and 5 units up to form $y = G(w)$, mapping $D(12, 33)$ to E. What is the sum of coordinates of E on $y = G(w)$?

a. 56

b. 50

c. 61

d. 63

31 **[Intermediate]** • **[MCQ]** • **Target: Function Evaluation, Composition, Transformations & Graphs**

In a model for the water treatment plant, in a calculation for the textile workshop (case 14), what is the minimum value of $f(x) = 1|x - 9|$?

a. -3

b. 0

c. -6

d. 6

32 **[Intermediate]** • **[MCQ]** • **Target: Function Evaluation, Composition, Transformations & Graphs**

In a model for the sports complex, in a calculation for the geology lab (case 15), what is the minimum value of $f(x) = 7|x + 5| - 10$?

a. -16

b. -8

c. -5

d. -10

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33 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the planetarium, in a calculation for the carpentry workshop (case 17), if $f(-7) = -18$, and $g(x) = f(x) - 3$, what is $g(-7)$?

a. -21

b. -12

c. -25

d. -16

34 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the regional airport shuttle, in a calculation for the historic lighthouse (case 18), if $f(8) = 11$, and $g(x) = f(x) + 5$, what is $g(8)$?

a. 8

b. 22

c. 16

d. 10

35 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the wind energy cooperative, consider the quadratic function $F(w) = 3w^2 - 15$ over the closed interval $[13, 18]$. What is the average rate of change $\Delta F/\Delta w$ on $[13, 18]$?

a. 96

b. 106

c. 93

d. 78

36 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the national park visitor center, a function F satisfies the identity $F(w - 26) = 11w + 6$ for all real w. What is the value $F(13)$?

a. 435

b. 441

c. 446

d. 409

37 [Intermediate] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the glassblowing studio, in a calculation for the neighborhood hardware store (case 22), what is the minimum value of $f(x) = 6|x + 7| + 3$?

a. -1

b. 11

c. 3

d. -4

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Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the maple syrup farm, let the functions R and S be defined by $R(m) = 7m - 9$ and $S(m) = m^2 + 12$. What is the composite value $R(S(9))$?

a. 642

b. 654

c. 633

d. 649

39 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the planetarium, $f(x) = \{-4x - 8, x < -5;\ -4x + 7, x \geq -5$ What is $f(-9)$?

a. 26

b. 28

c. 31

d. 21

40 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the wind energy cooperative, in a calculation for the community orchard (case 69), if $f(x) = 4x^2 + 6$ and $g(x) = 4x - 8$, what is $f(g(1))$?

a. 61

b. 77

c. 70

d. 74

41 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In a model for the marine research vessel, in the coordinate plane, the graph of $y = R(m)$ is translated 5 units right and 3 units up to form $y = S(m)$, mapping $M(17, 15)$ to N. What is the sum of coordinates of N on $y = S(m)$?

a. 35

b. 40

c. 43

d. 51

42 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The nonlinear function $h(x) = 5x^2 + c$, where c is a constant, models the output at the robotics club. If $h(3) = 67$, what is the value of $h(0)$?

a. 67

b. 27

c. 19

d. 22

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43 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

At the university lecture hall, the functions f and g are defined by $f(x) = 5x - 6$ and $g(x) = x^2 + 1$. What is the value of $f(g(4))$?

a. 15

b. 85

c. 79

d. 17

44 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

In the xy-plane, the graph of the quadratic function $y = f(x)$ has its vertex at $(3, 8)$. If $g(x) = f(x - 8)$ models the shifted curve at the culinary institute, what is the x-coordinate of the vertex of the graph of g?

a. 3

b. 11

c. -5

d. 16

45 [Challenging] • [MCQ] • Target: Function Evaluation, Composition, Transformations & Graphs

The nonlinear function $h(x) = 8x^2 + c$, where c is a constant, models the output at the wildlife sanctuary. If $h(2) = 57$, what is the value of $h(0)$?

a. 25

b. 57

c. 33

d. 23

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